Venkata Lakshmi Parimala Pasupuleti

Backend Engineer · Python · Django · Search Infrastructure · AI Stack · RAG Pipelines · Data Infrastructure · LangGraph · FastAPI · Open-Source AI

venkatalakshmiparimala@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Python backend engineer with 8+ years owning production data pipelines, search infrastructure, and distributed systems at scale. Built LeaseI — a hybrid TF-IDF + semantic search system with section-aware indexing, reranking, and an 8-agent LangGraph pipeline for knowledge retrieval over unstructured documents. Built AgentProbe — open-source AI evaluation infrastructure with CI-integrated regression harness and model routing across OpenAI/Anthropic/Groq/Ollama. Python mastery + Django in stack. Owns systems end-to-end: not just shipping features but designing data models, indexing strategies, and failure modes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; surfacing insights from raw business data into application-facing APIs.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern

- Built agentic backend system (LangGraph + FastAPI) with RAG over regulatory corpus, stateful orchestration, and validation guardrails — shipped to a real client on a small team with weekly iteration.

George Mason University Aug 2024 – Aug 2025

Research Associate

- Built multi-source data pipelines (Python, R) across 190+ countries; applied anomaly detection (Isolation Forest) to surface data quality issues; fine-tuned HuggingFace classification model with PyTorch, deployed inference API on Quart.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Owned data indexing and serving infrastructure for 7 years** — annual pipeline expanding few thousand federal source rows into ~180,000 domain records powering eligibility, rent, and financial calculations for 100K+ housing units. Owned schema design, indexing strategy, validation, rollback, and incident response. Zero failures across all release cycles.
- **Built RabbitMQ-based async processing system** for high-volume record updates — idempotency guarantees, retry logic, owned through production incidents. Designed for correctness under concurrent writes, not just throughput.
- **Optimized 200+ SQL queries (30% gain)** via execution plan analysis, indexing, partitioning, and stored procedure refactoring across multi-tenant production schemas. Owned complex data models for eligibility, financial, and document workflows.
- **Led AWS S3 migration** for document storage — inventoried, mapped, validated, confirmed correctness post-cutover; 100% PHI compliance, zero incidents. Python/Bash automation pipelines eliminated 80% of manual operational workload.
- Shipped 20+ full-stack modules across 120+ Agile sprints; led microservices redesign with Docker — 50% scalability improvement, 40% deployment time reduction.

PROJECTS

Leasel: Search + Knowledge Retrieval System HYBRID SEARCH LANGGRAPH

Python · LangGraph (8-agent) · TF-IDF + Semantic Search · FastAPI · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built hybrid TF-IDF + semantic search with section-aware indexing, reranking, and citation grounding over a corpus of 1,240 unstructured PDFs — designed the retrieval stack from scratch (no LangChain), tuned chunking and ranking strategies to maximize retrieval precision.
- 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing — each agent scoped to one job; knowledge composable across agents via shared evidence model.
- Eval: 93.6% precision, 87.9% F1 across 10 classification rules on 1,240 documents; pass@3 88.3%.

AgentProbe: Open-Source AI Evaluation Infrastructure OPEN SOURCE CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend with 5-metric eval engine, CI regression harness with golden datasets, model routing across OpenAI/Anthropic/Groq/Ollama, and offline + online evaluation loops. Not a demo — a CI-integrated system with test coverage and version-gated quality gates.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages: Python (8+ yrs), JavaScript, TypeScript, Java, SQL, R, Bash

Backend & Search: Django, FastAPI, REST APIs, Elasticsearch, Solr, PostgreSQL, Redis, RabbitMQ, Celery, Temporal, Apache Kafka, Microservices, Distributed Systems, Idempotency, Multi-Tenant Architecture

AI Stack: LangChain, LangGraph, RAG Pipelines, Hybrid Search, Semantic Search, TF-IDF, Reranking, Embeddings, ChromaDB, Qdrant, LLM Evaluation, LLM-as-Judge, Model Routing, Hallucination Detection, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, HuggingFace, PyTorch

Data Infrastructure: ETL, Data Pipelines, AWS (S3, EC2, RDS, Glue, Lambda), Azure Databricks, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), Prometheus, Grafana